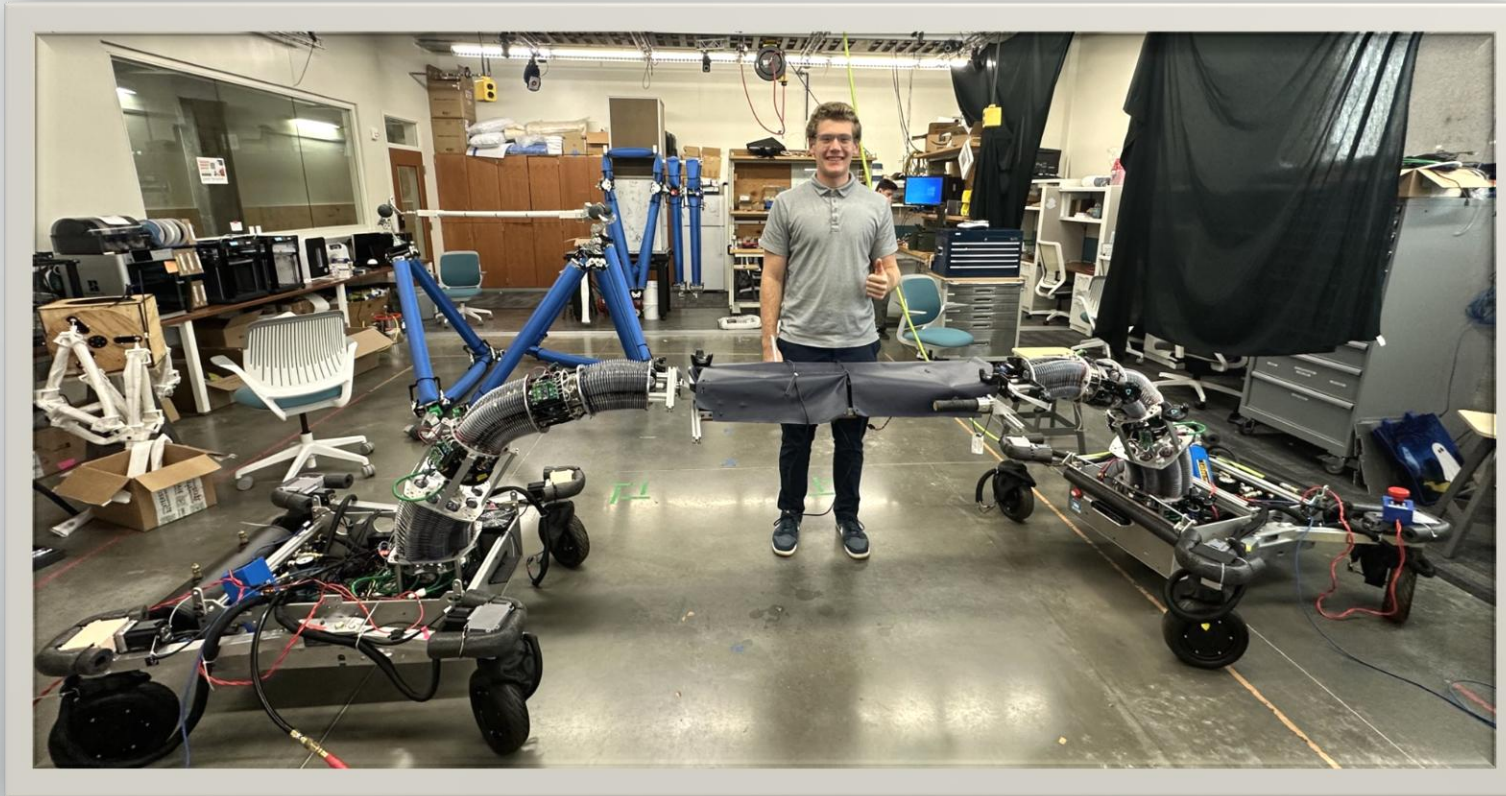


BYU Robotics and Dynamics Laboratory Dual Robot System Research Project



Project Mission and System Overview

- The BYU Robotics and Dynamics Lab is developing a robotic system to assist a human in collaboratively carrying a table.
- The system uses a pneumatic robotic arm instrumented with motion trackers to infer human intent
- Sensor feedback from the arm is used to generate real-time control commands for mobile base motion
- The original control architecture had only been implemented on a single robot platform
- Our objective was to deploy the existing behavior on a second robot, while improving robustness, modularity, and overall system behavior



Existing single-robot collaborative system
used as the baseline

Baseline System and Key Limitations



Second robot (Raksha) prior to software integration and hardware updates

- Planar translation had been implemented; no rotational dynamics or orientation control
- The codebase had only been implemented on a single robot
- Certain hardware for doing the table demo were only present on the first robot (Akela)
- Topics and nodes were not namespaced, preventing multi-robot deployment

Improvements Implemented

- Deployed and validated the existing code on the new robot (Raksha), establishing a consistent software baseline
- Implemented stabilization-focused rotational control logic to maintain stable robot orientation during motion
- Machined and integrated the hardware required to support the demonstration
- Implemented ROS namespacing to isolate topics, nodes, and parameters, enabling multi-robot operation

Results and Validation

- Successfully deployed the full control stack on both robots, enabling simultaneous operation
- Verified reliable pneumatic arm actuation on both robots; both arms performed as expected during testing and demonstration
- Demonstrated improved motion stability with rotational control enabled, while observing degraded behavior during reverse motion due to wheel orientation ambiguity
- Validated structural performance by successfully supporting the table load, confirming mechanical and control integration
- Encountered a hardware failure in one of Raksha's wheel motors that limited base mobility; calibration-based workarounds were attempted, but the issue prevented a fully successful end-to-end demonstration



Two-robot collaborative demonstration validating multi-robot control and mechanical integration

Next Steps and Improvements

- Diagnose and resolve the wheel motor failure on Raksha to restore full base mobility
- Refine and harden the rotational control logic to improve robustness across forward and reverse motion
- Extend the control framework to support commanded robot rotation; current rotation logic is stabilization-only and prevents drift rather than enabling active rotation
- Migrate the project to ROS 2 to improve scalability, maintainability, and long-term support
- Develop a simulation environment to enable faster iteration, debugging, and validation prior to hardware testing